

CLAIMIT

Every Bill is a Reward.

Rewards. Discounts. Cashbacks.

Full Platform Scope Document

Version 1.0

Tech Stack: React + Vite · FastAPI · MongoDB Atlas

1. Project Overview

Product Name: Claimit

Tagline: Every Bill is a Reward.

Problem Statement

Local businesses struggle to retain customers without expensive loyalty programs. Customers forget to return to shops they enjoyed. Advertisers have no cost-effective, hyperlocal ad platform to reach nearby shoppers. Sales affiliates (network marketers) have no structured digital tool to track shops they have onboarded. Claimit solves all three problems in one unified platform.

Target Users

Persona	Description	Pain Point
Shop Owner	Runs a local grocery, salon, gym, electronics store, etc.	Wants customer loyalty but cannot afford complex CRM systems
Advertiser / Brand	A digital marketing agency or brand wanting to push hyperlocal ads	Needs a targeted ad placement platform for local discovery apps
Sales Agent (Network Marketer)	Person who onboards shops to Claimit	Needs a dashboard to manage their shop portfolio and track earnings

Core Value Proposition

Claimit is a three-sided loyalty + advertising + affiliate platform. Shops pay annually to list and offer cashback rewards to customers. Advertisers pay per-ad-slot to reach local buyers through Claimit's consumer app. Sales agents earn commissions by onboarding shops. All three portals share the same auth infrastructure but have separate dashboards and workflows.

Monetization

- Shop annual subscription
- Advertising revenue: Home Page Banner, Promo Reelz, Brand Deals, Nearby Deals ad slots
- 1% shop incentive back to Claimit per transaction processed through the platform

2. User Roles

Three Portal Logins

The platform has one shared website with three distinct login entry points shown in the top navigation:

Button Label	Portal Name	Role
Login-Ads	Advertiser Portal	advertiser
Sales Login	Sales Agent Portal	sales_agent
Shop Login	Shop Owner Portal	shop_owner

All three share the same registration / OTP flow, but redirect to different dashboards post-login.

Permission Matrix

Feature	Advertiser	Sales Agent	Shop Owner	Super Admin
Create Ads	Yes	No	No	Yes
View Ad Analytics	Yes	No	No	Yes
Register Shop	No	Yes	Yes	Yes
Manage Offers	No	No	Yes	Yes
View Transaction History	Yes	Yes	Yes	Yes
View Earnings (team)	No	Yes	No	Yes
Add Team Members	No	Yes	No	Yes
Store Details Management	No	No	Yes	Yes
Ratings & Reviews	No	No	Yes	Yes
Approve Shops	No	No	No	Yes
Manage All Users	No	No	No	Yes

3. Authentication & Registration Flow

All three portals share the same registration and OTP verification flow. The portal type is passed as a role param (advertiser / sales_agent / shop_owner) during registration so the backend tags the user correctly.

3.1 Shared Header (All Pages)

Every page — authenticated or not — renders the top navigation bar:

- Left: Claimit logo (yellow circle + wordmark "claimit")
- Right nav links: Help Centre · Support · Terms of Service · Privacy Policy
- Auth buttons (unauthenticated): Login-Ads | Sales Login | Shop Login
- Auth state (authenticated): Shows user avatar/initials + dropdown (Profile, Logout)

3.2 Registration Page

Fields:

Field	Type	Required	Validation	Error
Phone Number	tel	Yes	10-digit Indian mobile	Enter a valid 10-digit number
Email Address	email	Yes	RFC 5321 email format	Enter a valid email address

Flow:

1. User lands on /register?portal=ads (or sales, shop)
2. Left panel: Illustrative image + motivational tagline (different per portal)
3. Right panel: "Register Your Account" form
4. On "Continue" click → POST /api/auth/register with {phone, email, role}
5. Backend: Check if email already exists → if yes, return 409 with "Account already exists, please login"
6. If new → create user record (isVerified: false, isApproved: false) → generate 6-digit OTP → store hashed OTP with 10-min TTL → send OTP via SMS + email
7. Redirect to OTP page
8. Bottom link: "Already have an account? login" → links to /login?portal=<portal>

3.3 OTP Verification Page

UI:

- 6 individual input boxes (auto-advance on digit entry, backspace goes back)
- "Didn't receive code? Resend OTP" link (available after 60s cooldown)
- "Verify OTP" button

Flow:

9. User enters 6 digits
10. POST /api/auth/verify-otp with {email, otp}

11. Backend: fetch hashed OTP → bcrypt.compare → if match and not expired → update user isVerified: true
12. If sales_agent or advertiser → skip to "Add Details" page or direct dashboard
13. If shop_owner → redirect to shop registration wizard

Edge Cases:

- Expired OTP (>10 min): 400 { detail: "OTP expired. Please request a new one." }
- Wrong OTP: 400 { detail: "Invalid OTP." } (max 5 attempts, then lock for 15 min)
- Resend: max 3 resends/hour per email, enforce via rate-limit middleware

3.4 Login Flow

Each login attempt triggers a fresh OTP to the registered phone + email. On successful OTP verify → issue JWT access token (15 min) + refresh token (7 days, httpOnly cookie). Role decoded from JWT → frontend redirects to appropriate dashboard.

3.5 Token Strategy

- Access Token: JWT, 15-min expiry, payload: { userId, role, email }
- Refresh Token: 7-day expiry, stored in httpOnly cookie
- All protected routes validate Bearer token via Depends(get_current_user) in FastAPI
- Token refresh endpoint: POST /api/auth/refresh

4. Onboarding Flow

Advertiser (Login-Ads)

After OTP verify → lands directly on Advertiser Dashboard with "Get Started" empty state showing "Post Advertising" card → click "Get Started" → redirects to "Create Ad" flow (Step 1: Choose Ad Type).

Sales Agent (Sales Login)

After OTP verify → prompted with "Add Your Details" form:

- Your Name (text, required)
- Address (text, required)
- Pincode (6-digit, required)
- Geo Location → "Use my current location" button (calls browser Geolocation API)
- Continue → lands on Sales Dashboard

Shop Owner (Shop Login)

After OTP verify → "Choose The Services" screen → one option card: "Add Shop or Business" → "Get Started" button → Shop Registration Wizard (multi-step).

5. Full Page-by-Page Breakdown

PORTAL A: ADVERTISER (Login-Ads)

Page A1: Advertiser Registration

Purpose: Register a new advertiser account. Entry via "Login-Ads" button in nav.

URL: /register?portal=ads

Left Panel:

- Illustration: Woman running up green arrows (growth theme)
- Tagline: "Maximize Your Revenue with Claimit"
- Sub-text: "Join the affiliate network where every sale generates extra value for your business and a cashback bonus for you."

Right Panel — Form:

- Title: "Register Your Account"
- Subtitle: "We'll send you a verification code to confirm your details"
- Phone Number field (tel input)
- Email Address field
- Continue button (blue, full-width)
- "Already have an account? login" link

Edge Cases:

- Empty fields → inline red error under each field
- Invalid email format → "Enter a valid email"
- Already registered → toast "Account exists. Please login."

Page A2: Advertiser OTP Verification

URL: /verify-otp?portal=ads

Left Panel: Illustration: Bar chart with people + upward red arrow (growth). Tagline: "Turn Every Bill Into a Reward for You"

Right Panel:

- Title: "Enter OTP"
- Subtitle: "We've sent a 6-digit code to your phone and email"
- 6 individual OTP boxes (auto-tab, styled with rounded borders)
- "Didn't receive code? Resend OTP" link (blue text)
- "Verify OTP" button (blue, full-width)

Page A3: Advertiser Dashboard (Empty State)

URL: /dashboard/ads

Sidebar (left panel, 280px):

- User avatar (pink circle, initials) + email + phone number
- Nav items: Dashboard (active), Create Ad, Transaction History, Help & Support

Main Content:

- Heading (blue): "Get Started Join and Grow with us!"
- Large center card with blue dashed border: AD icon + Label: "Post Advertising"
- "Get Started" button → navigates to /dashboard/ads/create

Page A4: Advertiser Dashboard (Active State)

URL: /dashboard/ads (post first ad creation)

Main Content:

- Title: "Dashboard" + subtitle: "Welcome to your advertising overview"
- "Create New Ad" button (top-right, blue)
- Stats Row: Total Ads | Total Views | Total Clicks
- "Your Advertisement" section with tab filter: All | Active | Scheduled
- Table rows: Ad Name | Publish Date | End Date | Views | Clicks | Status badge

Page A5: Create Ad — Step 1: Choose Ad Type

URL: /dashboard/ads/create/type

Content: 4 cards in a 2x2 grid, each with preview thumbnail, Ad Type name, description, price, and radio button.

Ad Type	Format	Price
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Home Page Banner Ad	Video / Static Thumbnail	Rs. 840
Promo Reelz Ad	Video / Static Thumbnail	Rs. 1400
Brand Deals Ad	Thumbnail + Details	Rs. 1400
Nearby Deals Ad	Thumbnail + Details	Rs. 1400

"Next" button (blue, bottom-right) — disabled until a selection is made.

Page A6: Create Ad — Step 2: Add Details

URL: /dashboard/ads/create/details

Layout: Two-panel — form on left, live preview on right.

Left Panel Form Sections:

- Ad Title (text input, required, max 80 chars)
- Ad Description (textarea, required, max 300 chars)
- Creative Upload: Drag-and-drop zone — JPG, PNG, MP4 (Max 50MB)
- Pincode (6-digit numeric, required)
- Price Summary card showing selected ad price including 18% GST
- "Continue" button → Step 3

Right Panel: Live preview phone frame (iPhone mockup) with Claimit consumer app. Preview renders the ad in the appropriate slot. Real-time update as user types.

Page A7: Create Ad — Step 3: Publish / Schedule

URL: /dashboard/ads/create/publish

Duration: All advertisements run for a fixed duration of 7 days.

Schedule Options:

- Publish Today — ad goes live immediately and runs for 7 days
- Schedule Ad — pick a future date (up to 10 days from today)

Validation:

- If Schedule Ad selected but no date: "Please select a publish date"
- Date cannot be more than 10 days from today
- Date cannot be in the past

Page A8: Finish Payment

URL: /dashboard/ads/create/payment

Content (centered card):

- Title: "Finish Your Payment" — Subtitle: "Publish your shop instantly."
- Blue price card: Price Summary with 18% GST included
- Payment methods: UPI (GPay, PhonePe, Paytm, BHIM) | Cards (Visa, Mastercard, Amex, RuPay) | Netbanking
- "Pay now" button (blue, full-width)

Page A9: Payment Success

URL: /dashboard/ads/create/success

- Green checkmark icon in green circle (animated pulse)
- Title: "Payment Successful!"
- Subtitle: "Your advertisement has been scheduled successfully"
- Ad Information card: Ad Type | Publish Date | Amount paid | Ad End Date
- Two buttons: "Create Another Ad" (blue) | "Go to Dashboard" (outline)

Page A10: Transaction History

URL: /dashboard/ads/transactions

Table columns: Date | Ad Type | Amount | Status | Invoice Download

Page A11: Help & Support

URL: /dashboard/ads/support

- FAQ accordion
- Contact form: Subject + Message + Submit

PORTAL B: SALES AGENT (Sales Login)

Page B1: Sales Agent Registration

URL: /register?portal=sales

Left Panel:

- Tagline: "Create Lasting Bonds With Every Bill"
- Sub-text: "Give your customers a reason to smile by providing points and cashback..."

Right Panel: Same form as Advertiser Registration (phone + email + Continue). Role: sales_agent

Page B2: OTP Verification

URL: /verify-otp?portal=sales

- Tagline: "Boost Your Profits with Every Transaction"
- Sub-text: "Benefit from our 1% shop incentive..."
- Right Panel: Same 6-box OTP form

Page B3: Sales Agent Onboarding — Add Details

URL: /onboarding/sales

Form Fields:

- Title: "Add Your details"
- Your Name (text, required)
- Address (text, required)
- Enter your Pincode (6-digit, required)
- Geo Location → "Use my current location" button (grey outline, calls Geolocation API)
- Continue button

Page B4: Sales Agent Dashboard

URL: /dashboard/sales

Sidebar:

- Avatar + Name + User ID (e.g., "Dhanush | User Id: 528882")
- Nav: Dashboard, Tutorial Zone, Your Team, Earning, Settings

Main Content:

- Title: "Dashboard" — Subtitle: "Welcome to Your overview"
- "Add Shop" button (top-right, blue)
- Stats Row: Total Shop | Total Revenue | Total Members
- "Your Shops" section with tabs: All | Active | Pending

Page B5: Tutorial Zone

URL: /dashboard/sales/tutorials

- Title: "Tutorials Zone"

- 2x2 grid of video placeholders — each card: thumbnail + play button overlay + video title

Page B6: Your Team

URL: /dashboard/sales/team

- List of team members with their stats
- "Invite Member" button → sends invitation link
- Member rows: Name | User ID | Total Shops | Total Revenue | Join Date

Page B7: Earning

URL: /dashboard/sales/earning

- Total Earnings card
- Withdrawal history table: Date | Shop Name | Amount | Status
- "Withdraw" button (enabled when minimum balance met)

Page B8: Sales Agent Settings

URL: /dashboard/sales/settings

- Personal info: Name, Email, Phone (editable)
- Change location (Pincode, Geo)
- Account security section

PORTAL C: SHOP OWNER (Shop Login)

Page C1: Shop Owner Registration

URL: /register?portal=shop

- Tagline: "Scale Your Business Like Never Before"
- Sub-text: "Join Claimit to turn every transaction into a growth opportunity..."
- Right Panel: Same phone + email form, role: shop_owner

Page C2: OTP Verification

URL: /verify-otp?portal=shop

- Tagline: "Take Your Business to the Next Level"
- Right Panel: 6-box OTP form

Page C3: Choose The Services (Post OTP)

URL: /onboarding/shop/services

Sidebar: Add your Shop or business, Dashboard, Analytics, Offer management, Store Details Management, Ratings & reviews, Settings

Main Content: Title: "Choose The Services" — Single card: shop emoji icon + "Add Shop Or Business" → "Get Started" button → navigates to Shop Registration Step 1

Page C4: Shop Registration — Step 1: Shop Details

URL: /onboarding/shop/register/details

Form Fields:

- Shop Name (text, required, max 100 chars)
- Shop Address (text, required, max 300 chars)
- Geo Location → "Use my current location" button (outline)
- About Shop (textarea, optional, max 500 chars)
- Continue button

Page C5: Shop Registration — Step 2: Upload Photos

URL: /onboarding/shop/register/photos

- Upload Cover Photo (required): JPG, PNG, max 5MB
- Upload Shop Photos (Optional): up to 5 files, JPG/PNG, max 5MB each
- Continue button

Page C6: Shop Registration — Step 3: Business Category

URL: /onboarding/shop/register/category

Dropdown categories: Groceries, Supermarket, Pharmacy, Bakery, Cafe, Salon, Gym, Electronics, Clothing, Restaurant, Jewellery, Hardware, Other

Page C7: Shop Registration — Step 4: Shop Type

URL: /onboarding/shop/register/type

Two radio cards:

- Reward Shop — Customers earn points/cashback when they pay bills at this shop
- Redeem Shop — Customers can use their accumulated points at this shop
- "Know more?" link (blue) → opens explanation modal

Page C8: Shop Registration — Step 5: Review & Submit

URL: /onboarding/shop/register/review

Review sections (each with Edit button navigating back to that step):

- Shop Details: Shop Name, Shop Address, Geo Location, Description
- Shop Photos: 4-photo thumbnail grid
- Category: category name + icon
- Shop Type: type + icon + discount %

Page C9: Shop Registration Payment

URL: /onboarding/shop/register/payment

- Annual subscription payment
- Payment method selection: UPI | Cards | Netbanking
- "Pay now" button

Page C10: Payment Success

URL: /onboarding/shop/register/success

- Green checkmark icon
- "Payment Successful!" message
- "Go to Dashboard" button → /dashboard/shop

Page C11: Shop Owner Dashboard

URL: /dashboard/shop

Sidebar:

- Shop avatar + Shop Name
- Nav: Dashboard, Offer management, Store Details Management, Ratings & reviews, Settings

Main Content:

- Stats Row: Total Reward Given | Total Redeem Used
- "Rewards / Redeem" toggle tabs
- Rewards tab columns: Customer | Reward Points | Bill Amount | Date
- Redeem tab columns: Customer | Redeemed (points used) | Bill Amount | Date

Page C12: Offer Management

URL: /dashboard/shop/offers

- Current offer display card (large text showing current %)
- Choose discount: 6 radio buttons — 10% | 15% | 20% | 25% | 30% | 35%
- "Publish Offer" button (blue, right-aligned)

Logic: When a customer scans their bill at this shop in the consumer app, this discount % is displayed and applied to reward calculation.

Page C13: Store Details Management

URL: /dashboard/shop/store-details

Each field is a row with current value + Edit button. Edit button opens inline edit mode with Save / Cancel:

- Shop Name
- Description
- Shop Address
- Shop Geo Location
- Category (dropdown reappears)
- Shop Type (radio cards reappear)

Page C14: Ratings & Reviews

URL: /dashboard/shop/reviews

Stats Row:

- Average Rating (star icon)
- Total Reviews (badge icon)
- Positive Reviews (thumbs-up icon, %)

Recent Reviews section with tab filter (All | Oldest). Each review card:

- User avatar (blue circle with initial)
- Name + Date
- Star rating (1–5 gold stars)
- Review text
- "Reply" link (blue) → opens inline text input to post owner reply

Page C15: Account Settings

URL: /dashboard/shop/settings

- Your Information: Email Address + Phone Number (editable)
- Annual Subscription section (blue card): Price Summary + Next renewal date

6. Dashboard Details

Advertiser Dashboard Widgets

Widget	Data Source	Update Frequency
Total Ads count	ads collection	Real-time
Total Views	ad_impressions collection	Every 30 min batch
Total Clicks	ad_clicks collection	Every 30 min batch
Ad List Table	ads collection	On page load
Status Badges	ad.status field	Real-time

Sales Agent Dashboard Widgets

Widget	Data	Update Frequency
Total Shop	shops where agentId = me	Real-time
Total Revenue	SUM(earnings)	Daily batch
Total Members	team.members.count	Real-time
Shop List	shops collection filtered by agent	Real-time

Shop Owner Dashboard Widgets

Widget	Data	Update Frequency
Total Reward Given	transactions where type=reward	Real-time
Total Redeem Used	transactions where type=redeem	Real-time
Transaction Table	transactions collection	Real-time

7. AI Features (Suggested)

While the current designs do not show explicit AI features, the following AI layers are recommended:

7.1 Smart Ad Targeting

- What: AI ranks nearby shops/brands in consumer app feed based on user history + location
- Model: Collaborative filtering on MongoDB transaction data
- Input: User location, purchase history, category preferences
- Output: Ranked list of deals shown in the consumer app

7.2 Offer Optimization Suggestions

- What: When Shop Owner is on Offer Management, AI suggests optimal discount %
- Model: Historical data analysis — e.g. shops with 20% discount saw more transactions
- Input: Shop category, local competitor discounts, historical redemption rate
- Output: Recommendation card suggesting best-performing discount

7.3 Ad Copy Assistant (Advertiser Portal)

- What: On "Add Details" page, a "Generate with AI" button auto-fills Ad Title and Description
- Model: Anthropic Claude API (claude-sonnet-4-6)
- Input: Ad type + pincode + business category
- Output: Suggested title + description, user can edit

7.4 Review Sentiment Analysis

- What: Shop owner dashboard shows sentiment score per review (Positive / Neutral / Negative badge)
- Model: Lightweight NLP classifier (VADER or Hugging Face)
- Input: Review text
- Output: Sentiment label + score

7.5 Fraud Detection

- What: Flag suspicious bill scan patterns (same user scanning same bill multiple times)
- Model: Rule-based anomaly detection + ML classifier
- Input: transaction.userId, transaction.shopId, transaction.timestamp, transaction.billAmount

- Output: Flag transaction for admin review

8. Admin Panel

URL: /admin (separate login, role: super_admin)

Admin Sidebar Nav

- Dashboard
- Users (All / Advertisers / Sales Agents / Shop Owners)
- Shops (All / Active / Pending / Suspended)
- Ads (All / Active / Scheduled / Rejected)
- Transactions
- Reports & Analytics
- Settings

Admin Capabilities

Feature	Description
User Management	View all users by role, search, filter by status, export CSV
Account Approval	Approve / reject shop registrations
Account Lock/Unlock	Lock user accounts with reason
Ad Moderation	Review submitted ads, approve/reject creative content
Shop Verification	Manual review of submitted shop details + photos
Analytics Overview	Total users, total revenue, daily active shops, ad impressions
Role Management	Promote sales agent to team lead
Audit Log	All admin actions logged: adminId + action + targetId + timestamp

Admin Auth

Admin uses password login (not OTP). All /api/admin/* routes require AdminGuard middleware.

9. Database Design (MongoDB Atlas)

Collection: users

- `_id`: ObjectId
- `email`: String (unique, indexed)
- `phone`: String (unique, indexed)
- `role`: Enum['advertiser', 'sales_agent', 'shop_owner', 'super_admin']
- `isVerified`: Boolean (default: false)
- `isApproved`: Boolean (default: false)
- `isActive`: Boolean (default: true)
- `name`, `address`, `pincode`, `geoLocation`: { lat, lng }
- `avatarUrl`, `onboardingComplete`: Boolean
- `createdAt`, `updatedAt`: Date
- Indexes: email, phone, role

Collection: otp_store

- `email`: String (indexed)
- `otpHash`: String
- `purpose`: Enum['register', 'login', 'reset']
- `expiresAt`: Date (TTL index, auto-delete after 10 min)
- `attempts`: Number (default: 0)

Collection: refresh_tokens

- `userId`: ObjectId (ref: users)
- `token`: String (hashed)
- `expiresAt`: Date
- `isRevoked`: Boolean (default: false)

Collection: shops

- `ownerId`, `agentId`: ObjectId (ref: users)
- `shopName`, `shopAddress`, `geoLocation`: { lat, lng }, `pincode`
- `aboutShop`, `coverPhotoUrl`, `shopPhotoUrls`: [String]
- `category`, `shopType`: Enum['reward', 'redeem']
- `discountOffer`: Number (default: 10)
- `status`: Enum['draft', 'pending_approval', 'active', 'rejected', 'suspended']
- `isPaid`: Boolean, `subscriptionStart`, `subscriptionEnd`: Date
- `averageRating`: Float, `totalReviews`: Number

- Indexes: ownerId, agentId, pincode, category, status, geoLocation (2dsphere)

Collection: ads

- advertiserId: ObjectId (ref: users)
- adType: Enum['home_banner', 'promo_reelz', 'brand_deals', 'nearby_deals']
- title, description, creativeUrl, pincode
- publishOption: Enum['today', 'scheduled'], publishDate, endDate: Date
- status: Enum['draft', 'pending_payment', 'scheduled', 'active', 'expired', 'rejected']
- isPaid: Boolean, paymentId: String
- views, clicks: Number (default: 0)
- Indexes: advertiserId, status, publishDate, pincode

Collection: transactions

- shopId: ObjectId (ref: shops)
- customerId, customerName: String
- type: Enum['reward', 'redeem']
- billAmount, rewardPoints, redeemedPoints: Number
- date, createdAt: Date
- Indexes: shopId, date, type

Collection: reviews

- shopId: ObjectId (ref: shops)
- customerId, customerName: String
- rating: Number (1-5), text: String
- ownerReply: String
- sentiment: Enum['positive', 'neutral', 'negative']
- Indexes: shopId, rating

Collection: sales_earnings

- agentId: ObjectId (ref: users)
- shopId: ObjectId (ref: shops)
- amount: Number
- status: Enum['pending', 'paid'], paidAt: Date

Collection: audit_logs

- adminId: ObjectId (ref: users)
- action: String
- targetType: Enum['user', 'shop', 'ad']
- targetId: ObjectId
- details: Object, createdAt: Date

Collection: tutorials

- title, videoUrl, thumbnailUrl: String
- duration: Number, order: Number

10. Data Flow Diagrams

Flow 1: User Registration & OTP Verification

14. User fills phone + email, clicks Continue
15. Frontend → POST /api/auth/register {phone, email, role}
16. FastAPI checks if email exists in MongoDB
17. If new → Insert user {isVerified:false}
18. Generate 6-digit OTP, bcrypt hash → Insert to otp_store {otpHash, expiresAt: +10min}
19. Send OTP via email + SMS
20. Frontend redirects to /verify-otp
21. User enters 6-digit OTP → POST /api/auth/verify-otp {email, otp}
22. FastAPI fetches otp_store → bcrypt.compare → Update user isVerified:true
23. Return {accessToken, role} → Frontend redirects to portal dashboard

Flow 2: Shop Registration & Admin Approval

24. Shop Owner completes 5-step shop wizard
25. POST /api/shop/register/step1 → Create shop {status: draft}
26. POST /api/shop/register/step2 → Upload images to Cloudinary → Store URLs
27. PATCH step3 (category) + step4 (type)
28. POST /api/shop/register/submit → Set shop.status: pending_approval → Notify admin
29. Admin reviews and PATCH /api/admin/shops/:id/approve
30. Set shop.status: active → Email ShopOwner "Your shop is approved!"

Flow 3: Ad Creation & Payment

31. Advertiser selects Ad Type
32. Fills details + uploads creative → POST /api/advertiser/ads/draft → Store draft
33. Chooses publish option → PATCH /api/advertiser/ads/:id/publish
34. Selects payment method + pays → POST /api/advertiser/ads/:id/payment
35. FastAPI processes payment → Set ad.isPaid:true, status: scheduled/active
36. Return payment success to frontend

Flow 4: Token Refresh

37. Frontend sends expired accessToken → FastAPI returns 401 Unauthorized
38. Frontend → POST /api/auth/refresh (refreshToken in httpOnly cookie)
39. FastAPI validates refreshToken, checks isRevoked → Issues new accessToken
40. Frontend retries protected route with new token

11. API Design

Auth Endpoints (All Portals)

Method	Endpoint	Auth	Description
POST	/api/auth/register	Public	Register new user
POST	/api/auth/verify-otp	Public	Verify OTP, get token
POST	/api/auth/login	Public	Initiate login
POST	/api/auth/verify-login-otp	Public	Complete login
POST	/api/auth/refresh	Cookie	Refresh access token
POST	/api/auth/logout	JWT	Revoke refresh token
POST	/api/auth/resend-otp	Public	Resend OTP

Advertiser Endpoints

Method	Endpoint	Auth	Description
GET	/api/advertiser/me	JWT	Get advertiser profile
GET	/api/advertiser/ad-types	JWT	List ad types + pricing
GET	/api/advertiser/ads	JWT	Get all my ads

GET	/api/advertiser/stats	JWT	Dashboard stats
POST	/api/advertiser/ads/draft	JWT	Create ad draft
PATCH	/api/advertiser/ads/:id/publish	JWT	Set publish schedule
POST	/api/advertiser/ads/:id/payment	JWT	Complete payment
GET	/api/advertiser/ads/:id/receipt	JWT	Get payment receipt
GET	/api/advertiser/transactions	JWT	Transaction history

Sales Agent Endpoints

Method	Endpoint	Auth	Description
GET	/api/sales/me	JWT	Get agent profile
PATCH	/api/sales/onboarding	JWT	Complete onboarding
GET	/api/sales/dashboard	JWT	Dashboard stats
GET	/api/sales/shops	JWT	My onboarded shops
GET	/api/sales/team	JWT	Team members
GET	/api/sales/earnings	JWT	Earnings history
GET	/api/sales/tutorials	JWT	Tutorial videos

Shop Owner Endpoints

Method	Endpoint	Auth	Description
GET	/api/shop/me	JWT	Get shop profile
POST	/api/shop/register/step1	JWT	Register step 1
POST	/api/shop/register/step2	JWT	Upload photos
PATCH	/api/shop/register/step3	JWT	Set category
PATCH	/api/shop/register/step4	JWT	Set shop type
POST	/api/shop/register/submit	JWT	Submit for review
POST	/api/shop/payment	JWT	Pay subscription
GET	/api/shop/dashboard	JWT	Dashboard stats
GET	/api/shop/transactions	JWT	Transaction list
GET/PATCH	/api/shop/offer	JWT	Get/update offer
GET/PATCH	/api/shop/details	JWT	Get/edit store details

GET	/api/shop/reviews	JWT	Get reviews
POST	/api/shop/reviews/:id/reply	JWT	Reply to review
GET	/api/shop/subscription	JWT	Subscription info
PATCH	/api/shop/account	JWT	Update account

Admin Endpoints

Method	Endpoint	Auth	Description
POST	/api/admin/auth/login	Public	Admin login
GET	/api/admin/users	Admin JWT	List all users
PATCH	/api/admin/users/:id/approve	Admin JWT	Approve user
PATCH	/api/admin/users/:id/lock	Admin JWT	Lock account
GET	/api/admin/shops	Admin JWT	List shops
PATCH	/api/admin/shops/:id/approve	Admin JWT	Approve shop
PATCH	/api/admin/shops/:id/reject	Admin JWT	Reject shop
GET	/api/admin/ads	Admin JWT	List all ads
PATCH	/api/admin/ads/:id/approve	Admin JWT	Approve ad
GET	/api/admin/analytics	Admin JWT	Platform analytics
GET	/api/admin/audit-logs	Admin JWT	Audit log

12. Security Considerations

Area	Implementation
OTP Storage	bcrypt hash (rounds=12), TTL index auto-delete after 10 min
OTP Attempts	Max 5 wrong attempts → lock OTP for 15 min, return 429

JWT Access Token	15-min expiry, HS256, payload: {userId, role, email}
JWT Refresh Token	7-day expiry, stored as httpOnly cookie, hashed in DB
Password (Admin)	bcrypt hash (rounds=12), never plaintext
RBAC	FastAPI Depends() guards: get_current_user + role check on every protected route
Rate Limiting	slowapi: /register: 3/hour/IP; /login: 5/15min/IP; /resend-otp: 3/hour/email
Input Sanitization	Pydantic v2 validators on all request bodies (EmailStr, constr, regex)
File Upload	Max 50MB ads creative, 5MB photos; validate MIME type server-side
CORS	Allow only registered frontend origin (Vite dev + production domain)
HTTPS	Enforce in production; 301 redirect HTTP to HTTPS
Admin 2FA	TOTP via pyotp (Google Authenticator compatible)
Audit Logging	Every admin action → insert to audit_logs collection
XSS	React auto-escapes; no dangerouslySetInnerHTML used
MongoDB Injection	Motor/Beanie with Pydantic models prevents query injection

13. Edge Cases & Validations

Scenario	Handling
Empty required fields	Inline red error under field; button remains disabled
Invalid email format	"Enter a valid email address" — client-side + server-side
Duplicate email registration	409: "An account with this email already exists. Please login."
OTP expired	410: "OTP has expired. Click Resend OTP." — old OTP deleted from DB
Wrong OTP (< 5 attempts)	400: "Invalid OTP. X attempts remaining."
Wrong OTP (5th attempt)	429: "Too many attempts. Please request a new OTP in 15 minutes."
Login before OTP verify	403: "Please verify your OTP first."
Shop login before admin approval	403: "Your shop registration is under review."
Resend OTP limit (>3/hour)	429: "Too many resend attempts. Try again in 1 hour."

Ad creative > 50MB	Client-side: "File exceeds 50MB limit" before upload
Invalid pincode	"Enter a valid 6-digit pincode" — regex: /^\d{6}\$/
Schedule date > 10 days	Date picker disables those dates; API validates: 400 if exceeded
Schedule date in the past	Date picker disables past dates
Accessing other user's resources	403: "Forbidden" — all queries filter by userId from JWT
Expired JWT on protected route	401 → frontend catches → attempts refresh → if refresh expired → redirect to login
File wrong format	"Only JPG, PNG, and MP4 files are supported"
Network failure during payment	Show "Payment failed. Please try again." — no ad record until payment confirmed
Admin approves already-active shop	409: "Shop is already active"
DB connection failure	503: "Service temporarily unavailable. Please try again later."
Concurrent OTP requests	Last OTP wins; previous OTP invalidated on new generation

14. Future Enhancements

Phase 2 Features

- Consumer App (React Native): The customer-facing app showing Nearby Deals, Brand Deals, Promo Reelz, Home Page Banners
- Bill Scan / QR Code: Shop generates QR; customer scans to log transaction and earn/redeem points
- Push Notifications: Notify customers of new deals from shops they've visited
- Referral System: Sales agents get referral links; tracks shop onboarding to agent

Scaling

- Redis: Cache hot queries (nearby deals by pincode, ad impressions counters)
- CDN: Cloudflare for serving ad creatives and shop photos
- Horizontal Scaling: FastAPI behind Unicorn + Nginx; containerized via Docker
- MongoDB Atlas Search: Full-text search for shop discovery

Mobile App

- React Native for iOS + Android (consumer app)
- React Native for shop owner: bill scan, offer management
- Expo EAS Build for CI/CD

Third-Party Integrations

- Razorpay / PhonePe: Production payment gateway (replace mock)
- MSG91 / Twilio: Production SMS OTP delivery
- SendGrid: Transactional email (OTP, receipts, approval notifications)
- Firebase: Push notifications for consumer app
- Google Maps API: Geolocation + nearby shop discovery map

Advanced AI

- Hyperlocal Deal Recommendation: Vector embeddings of shop categories + user preferences
- Dynamic Pricing for Ads: AI-suggested ad pricing based on demand in pincode
- Automated Ad Moderation: Claude/GPT to review ad creatives for policy compliance

Enterprise

- Multi-language Support: Tamil, Hindi, English (i18next)
- White-label: Franchisee portals with custom branding
- Analytics Dashboard v2: Export to CSV/Excel, date range filters, charts (Recharts)

Tech Stack Summary

Layer	Technology	Notes
Frontend	React 18 + Vite	React Router v6, code-split by portal
UI Library	Material UI v5	Custom theme: primary blue (#1565C0), Claimit brand
State Management	Zustand	Lightweight, portal-scoped stores
Backend	FastAPI (Python 3.11+)	Async, Pydantic v2
Database	MongoDB Atlas	Motor (async driver) + Beanie ODM
Auth	JWT (python-jose) + bcrypt	httpOnly refresh token cookie

OTP	6-digit + bcrypt hash	TTL index in MongoDB
File Storage	Cloudinary	Ad creatives + shop photos
Email	SendGrid	OTP, approvals, receipts
SMS	MSG91	OTP delivery
Rate Limiting	slowapi (FastAPI)	IP + email-based limits
Payments	Razorpay	UPI, Cards, Netbanking
Deployment FE	Vercel	Auto-deploy from main branch
Deployment BE	Railway or Render	Docker container
Admin Panel	Custom React (same codebase)	Role-gated routes
AI Layer	Anthropic Claude API	Ad copy generation, sentiment analysis